
The Pro-Action Operator: A Scientifically-Informed, Multi-Timescale Self-Regulation Model for LLM-Agents

Anonymous Author(s)

Affiliation

Address

email

Abstract

Contemporary LLM-agent frameworks — ReAct, LangGraph, OpenClaw — have prioritized *capability* over *motivation*, deciding *when* an agent acts through exogenous flow control or task-backlog heartbeats. Homeostatic reinforcement learning (Keramati–Gutkin 2014) and active-inference agents (Friston 2017; Tschantz et al. 2020) internalize regulation, but often through a compact objective rather than an explicit set of delayed regulatory subsystems. However, biological regulation is not single-timescale: converging work in stress physiology, self-regulation, affect, and cognitive control treats adaptation as the interaction of subsystems — attention, perception, hormonal (SAM adrenergic, seconds; HPA cortisol, minutes), emotional, neuropsychological, cognitive — exchanging signals across distinct delays, consistent with the mechanism behind Yerkes–Dodson inversion and delayed reappraisal (Porcelli–Delgado 2017). This work proposes the **Pro-Action operator** Γ , a recursively composable, multi-timescale thermostat system. Each subsystem k updates as

$$x_{k,t+1} = x_{k,t} - \kappa_k(x_{k,t} - x_{k,t}^*) + \lambda_k - \alpha_k \rho_k(a_t, e_{t+1}) + [W \phi(\mathbf{x}_{t-\tau_k})]_k,$$

with $\phi = \tanh$ elementwise, subsystem-specific delay τ_k encoding a dimensionless SAM–HPA-like ratio ($\tau_{\mathcal{H}}^{\text{HPA}} \gg \tau_{\mathcal{A}}$), and action selected by an arousal-gated mixer. Γ offers a finite, inspectable parameterization of regulation: the τ_k are explicit hyperparameters rather than hidden in orchestration depth. We specify three falsifiable properties a priori: (P1) elastic return to set-point; (P2) Yerkes–Dodson inverted-U; (P3) delayed reappraisal. As a Concept & Feasibility contribution, the paper reports the model, a lightweight falsifiability protocol, computational sanity checks, and a completed LLM-agent benchmark. The completed benchmark shows that the operator produces opponent-discriminative behavior that exogenous-activation baselines (ReAct) and structurally similar reductions (Random- Γ , Collapse-NC) do not — a feasibility claim, not a causal claim about the delay structure specifically.

Keywords. Homeostatic reinforcement learning; LLM agents; active inference; multi-timescale self-regulation; endogenous activation; cybernetics; self-regulation model.

1 Introduction

The deployment of LLM-based autonomous agents into settings of institutional consequence — health triage, financial advice, public deliberation — is moving faster than the science of how such agents should *regulate themselves*. Whether an agent speaks, waits, defers, or escalates tends to be decided outside the agent, by an orchestrator whose control rules may be less expressive than the

33 situations they coordinate: a small set of graph transitions, queues, or heartbeats is used to govern
34 high-dimensional and shifting social or task states. In Ashby’s terms, the controller may lack the
35 requisite variety needed to match the variety of the environment [1].

36 A frequently reported symptom, consistent with this variety mismatch, is the loss of activation
37 coherence in adversarial multi-agent deliberation: floor monopolization, context collapse, and
38 interventions that appear decoupled from any internal state the agent could be said to carry (evidence
39 observed by the authors in a companion work, currently under review). The pattern suggests that
40 *when* an agent acts may depend on structure the dominant paradigm does not explicitly represent.

41 Contemporary LLM-agent frameworks — ReAct, LangGraph, OpenClaw — have prioritized *ca-*
42 *pability* (what an agent can do) over *motivation* (why and when it should act at all), addressing
43 activation via cognitive while-loops and exogenous flow control. Homeostatic reinforcement learning
44 (Keramati–Gutkin 2014) and active-inference agents (Friston 2017; Tschantz et al. 2020) internalize
45 regulation, but do not provide a compact LLM-agent operator in which attention, interoception,
46 emotion, memory, cognition, and stress-like physiology are all explicit delayed regulators. Intelligent
47 animals, including humans, regulate behavior through multiple interacting systems — attention,
48 arousal, affect, memory, deliberation — operating on distinct timescales whose relative priority
49 shifts with context; dominant LLM-agent paradigms treat activation as a single-objective signal or
50 exogenous flow control.

51 The problem is older than LLM-agents. Mid-20th century behaviorism defined it precisely by
52 bracketing it: the stimulus–response mapping could be studied scientifically because the intervening
53 machinery — the *black box* — was deliberately left unspecified. Decades of subsequent research
54 opened that box from different angles: behavioral biology established that animal regulation is
55 multi-drive and hierarchical; stress physiology revealed fast (adrenergic, seconds) and slow (cortisol,
56 minutes) adaptation axes with distinct roles; affective neuroscience showed that emotion is not
57 an output label but a construction that shapes perception, memory, and action; evolutionary game
58 theory demonstrated that social behavior is governed by reciprocity and temporal discounting, not
59 instantaneous utility; cybernetics formalized why a regulator must match the variety of what it
60 controls. Each tradition contributed a piece. Nobody has assembled these threads for LLM-agents
61 into a single, compact, falsifiable operator. Section 2 reviews each contribution in detail. Opening
62 the black box matters beyond mechanistic completeness: (i) Γ provides a finite, inspectable state
63 vector, making each subsystem’s contribution auditable rather than buried in prompt engineering;
64 (ii) a controller tracking regulatory state creates the precondition for agents that adapt to their own
65 arousal trajectory; (iii) explicit emotional and hormonal channels provide the computational substrate
66 for empathy modeling — recognizing a counterpart’s affective state — which is structurally distinct
67 from fine-tuning on empathy datasets.

68 What remains comparatively unexplored is a single, computable, falsifiable operator that integrates
69 these disciplinary insights for LLM agents. The central biological clue is that regulation is not a
70 single controller but a *system of subsystems* — attention, perception, hormonal (SAM adrenergic,
71 seconds; HPA cortisol, minutes), emotional, neuropsychological, cognitive — exchanging signals
72 across distinct delays. The SAM–HPA timescale split is not decorative: it is consistent with the
73 mechanism behind Yerkes–Dodson inversion, post-stress reappraisal windows, and delayed-valuation
74 shifts [2]. The scientific question is therefore whether activation itself can be treated as a regulated
75 internal process — a coupled set of thermostat-like variables whose delayed interactions determine
76 when the agent should speak, wait, defer, or escalate — and whether the smallest such structure
77 remains distinguishable from a single-timescale regulator.

78 We propose the **Pro-Action operator** Γ as a model, not an experimental technique. Γ treats activation
79 as a recursively composable system of six coupled thermostats with explicit update delays. The
80 contribution type is Concept & Feasibility: the idea is intentionally broader than what can be validated
81 in a single benchmark, but it must still be made operational enough to fail. The contributions of
82 this paper are primarily *modeling*: (i) a formal operator integrating multi-timescale regulation; (ii)
83 the framing of Γ as a Self-Regulation Model (SRM) class complementary to capability-oriented
84 LAMs; (iii) an a priori specified falsifiability protocol; (iv) a reference implementation checked by
85 computational sanity tests; and (v) a completed 920-cell LLM-agent benchmark used as feasibility
86 evidence on opponent-discriminative behavior, with full causal attribution and BFI adjudication
87 remaining open.

88 2 Related work

89 **LLM-agent orchestration.** Recent LLM-agent frameworks such as ReAct [3], LangGraph, and
90 OpenClaw made language models operational by embedding them in action loops, tool graphs,
91 and multi-agent workflows. This was a necessary engineering step: without explicit structure, an
92 LLM has no durable procedure for acting in an environment. More recently, agent architectures
93 have incorporated persistent memory streams, reflection, and simulated affective states to enrich the
94 content of decisions [4–7]. The limitation common to both generations is that activation is usually
95 externalized: the graph or supervisor decides when the model acts; the model supplies content for
96 that externally selected moment. Γ is complementary: those works provide richer *content*; Γ provides
97 a principled internal account of *timing*.

98 **Internal regulation across timescales.** A different tradition begins from the opposite assumption:
99 action should be driven by an internal regulatory state. Homeostatic RL makes internal deficit part of
100 the reward [8]; active inference gives a broader variational formulation [9, 10]. These approaches
101 are important — they move regulation inside the agent — but do not yield a compact operator
102 with separately inspectable subsystem delays. Once regulation is internalized, the next question is
103 whether a single regulatory signal is sufficient. Stress physiology and allostasis suggest it is not: fast
104 sympathetic responses (SAM, seconds) and slower endocrine/cognitive recovery (HPA, minutes) play
105 distinct adaptive roles [11–13]. The remaining question is which internal variables deserve to be
106 modeled. Interoceptive and somatic-marker accounts connect bodily state to action readiness [14, 15];
107 affect-regulation accounts treat emotion as part of how situations are appraised [16?]; evolutionary
108 game theory adds reciprocity and temporal discounting as social constraints [18, 19]. Together, these
109 literatures motivate attention, interoception, emotion, memory, cognition, and hormonal dynamics as
110 functional regulators rather than decorative modules.

111 **Systems, cybernetics, and the gap.** If these variables are treated as separate regulators, the final
112 problem is structural: how should they be composed? Cybernetics supplies the relevant language:
113 Ashby’s law of requisite variety requires a regulator rich enough to match the environment [1]; Beer’s
114 Viable System Model adds recursive organization [20]; systemism frames the relevant object as
115 composition rather than isolated component [21, 22]. The gap might be narrow yet important. To
116 our knowledge, the threads above — exogenous orchestration, homeostatic internalization, multi-
117 timescale physiology, affect regulation, and cybernetic composition — have not been combined into a
118 small, falsifiable LLM-agent operator in which the six regulatory domains are explicit state variables
119 with explicit delays. That is the gap Γ is designed to occupy.

120 3 The Pro-Action operator

121 3.1 Axioms of homeostatic autonomy (A1–A9)

122 Before defining Γ , we make explicit the modeling commitments that the operator assumes. The
123 axioms are not empirical results or theorems; they state, in this model, what counts as an autonomous,
124 self-regulating agent, which internal variables are allowed to regulate action, and how those variables
125 may be composed. This is meant to avoid leaving the agent’s ontology implicit in implementation
126 choices. The axiom set was subjected to a formal coherence check using SMT-based tools (Z3) to
127 verify that no pair of axioms is mutually contradictory under the intended semantics; all checks
128 passed.

129 We extend the Axioms of Homeostatic Autonomy from the companion work (A1–A3) with six
130 additional axioms that make the multi-timescale structure explicit:

131 **A1–A3 (Kernel).** Autonomy, Impulse, and Quality Gate: the agent maintains an internal deficit δ_i
132 that drives action probability $p_i(t) = \sigma(D(\delta_i(t)))$, reduced by action quality g .

133 **A4 (Perception as initial appraisal).** Every stimulus is appraised before selective attention. A
134 perception operator $\mathcal{P}$ produces an initial situational appraisal $\tilde{o}_t = \mathcal{P}(o_t, \mathcal{I}_t, \mathcal{M}_t; x_{\mathcal{P},t})$, encoding
135 what the stimulus means independent of current priorities.

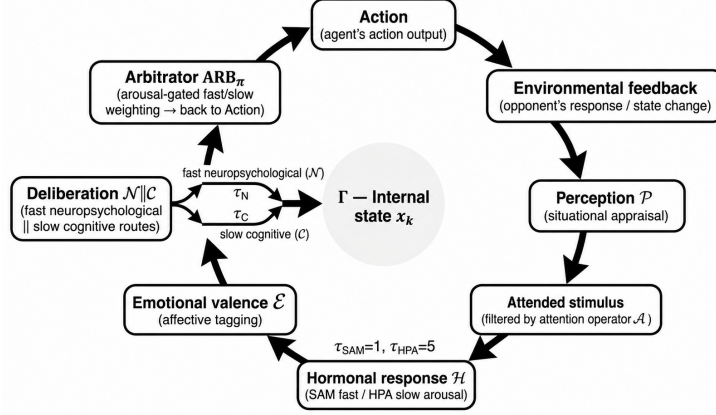


Figure 1: The Pro-Action operator Γ : regulatory cycle in canonical clockwise order. Perception ($\mathcal{P}$, situational appraisal) feeds Attention ($\mathcal{A}$, saliency filtering), which gates the Hormonal arousal module ($\mathcal{H}$, SAM fast / HPA slow), Emotional valence ($\mathcal{E}$), and a Deliberation gate routing to the Neuropsychological fast path ($\mathcal{N}$, $\tau_{\mathcal{N}}$) or Cognitive slow path ($\mathcal{C}$, $\tau_{\mathcal{C}}$); the Arbitrator ARB_{π} applies arousal-gated weighting before action emission. Notation: $\circ$ denotes composition, $\parallel$ parallel evaluation (routes merged by ARB_{π}), and $\mathcal{O}[\cdot]$ the output projection selecting the final action.

136 **A5 (Attention as saliency gate).** Not every appraised feature enters the action-selection context. An
 137 attention operator $\mathcal{A}$ filters the perceived signal, selecting $o'_t = \mathcal{A}(\tilde{o}_t, \mathcal{D}_t, \mathcal{M}_t; x_{\mathcal{A},t})$ based on current
 138 drives and memory. Interoceptive signals $\mathcal{I}_t$ feed in parallel to both $\mathcal{P}$ and $\mathcal{A}$.

139 **A6 (Memory as ubiquitous prior).** A state $\mathcal{M}_t$ biases all subsystems and updates post-action.

140 **A7 (Drives as set-point generators).** Set-points are not constants: $\mathbf{x}_t^* = \mathcal{D}_t$.

141 **A8 (Fast/slow parallelism with arbitrator).** Two routes produce action candidates $c^{\text{fast}}, c^{\text{slow}}$; an
 142 arbitrator assigns precision weights $\pi_{\text{fast}}, \pi_{\text{slow}}$.

143 3.2 The operator

144 We define Γ as the composition (Figure 1):

$$\Gamma := \mathcal{O} \left[\text{ARB}_{\pi} \left(\underbrace{(\mathcal{N}_{\text{fast}} \circ \mathcal{H}_{\text{SAM}})}_{\text{fast route}} \parallel \underbrace{(\mathcal{C} \circ \mathcal{E} \circ \mathcal{A} \circ \mathcal{P})}_{\text{slow route}} \right) \right] \Big|_{\mathcal{I}, \mathcal{M}, \mathcal{D}} \quad (1)$$

145 Each subsystem k updates as:

$$x_{k,t+1} = x_{k,t} - \kappa_k(x_{k,t} - x_{k,t}^*) + \lambda_k - \alpha_k \rho_k(a_t, e_{t+1}) + [W \phi(\mathbf{x}_{t-\tau_k})]_k \quad (2)$$

146 with $\phi = \tanh$ elementwise. The delays τ_k are dimensionless non-negative integers (in agent-
 147 interaction rounds), not literal biological durations; what the model inherits from SAM-HPA physi-
 148 ology is the *ratio* $\tau_{\mathcal{H}}^{\text{HPA}} \gg \tau_{\mathcal{A}}$, not the absolute timescales. Action is selected by:

$$a_{t+1}^* = \arg \min_a \pi_f \mathcal{F}_{\text{short}} + (1 - \pi_f) \mathcal{F}_{\text{long}} + \gamma \|Q^{1/2}(\mathbf{x}_{t+1}(a) - \mathbf{x}_{t+1}^*)\|^2, \quad \pi_f = \sigma(c \cdot x_{\mathcal{H}} + b) \quad (3)$$

149 where $\mathbf{x}_{t+1}(a)$ denotes the predicted next state under candidate action a , depending on a through the
 150 action-quality term $\rho_k(a, e_{t+1})$ in Eq. (2) — it is this dependence that makes the homeostatic deficit
 151 penalty action-relevant. $\mathcal{F}_{\text{short}}, \mathcal{F}_{\text{long}}$ are generic cost functionals inspired by expected-free-energy
 152 control over horizons $h_s \ll h_l$, and $Q \succeq 0$ is a diagonal precision matrix (distinct from the coupling
 153 matrix W in Eq. (2)). In the IPD instantiation below we do not claim a strict active-inference
 154 implementation with an explicit generative model. The mixer is not literally executed as an argmin
 155 over LLM completions: the controller modulates the prompt seen by the LLM (Section 4.3) and the
 156 LLM emits the action; Eq. (3) is the compact controller-level expression of fast/slow arbitration that
 157 the prompt encodes.

3.3 Non-absorption of delay parameters

Proposition 1 (Non-absorption of delays). *Let Γ_τ denote Eq. (2) with delays $\tau = (\tau_1, \dots, \tau_6) \in \mathbb{Z}_{\geq 0}^6$ and coupling matrix $W \neq \mathbf{0}$. For any two delay vectors $\tau \neq \tau'$ with $\tau_k \neq \tau'_k$ for at least one k such that $[W]_{k,\cdot} \neq \mathbf{0}$, there is no reparametrization $(\kappa'_k, \lambda'_k, \alpha'_k)$ of the non-delayed terms that makes the impulse-response of $\Gamma_{\tau'}$ identical to that of Γ_τ . Equivalently, τ_k is not absorbable into $(\kappa_k, \lambda_k, \alpha_k)$.*

The argument (Appendix A) is an order-mismatch in the discrete-time z-domain: a delayed subsystem has transfer-function denominator degree $\tau_k + 1$, enforcing τ_k zero-output samples by causality that no delay-free reparametrization can replicate. This establishes that τ_k is identifiable from observed trajectories and that Γ occupies a distinct model class from any delay-free parametrization of the same dimension.

3.4 Reductions to simpler models

Reduction to Driveplexity. Driveplexity (companion work, currently under review) is a single-drive homeostatic activation model in which each LLM-agent maintains a scalar epistemic deficit δ_i that decays at a fixed rate and is reduced by the quality of each action executed. Γ subsumes it: when $x_{\mathcal{A}} = \dots = x_{\mathcal{N}} \equiv 0$, $\mathcal{A} = \text{id}$, $\mathcal{I} = \mathcal{M} = \mathcal{D} = \emptyset$, $\pi_{\text{fast}} \equiv 0$, $W = \mathbf{0}$, $\kappa_{\mathcal{C}} = 0$, and $\mathbf{x} \rightarrow x_{\mathcal{C}} \equiv \delta_i$, Equation (2) reduces to $\delta_{t+1} = \delta_t + \lambda - \alpha \cdot g(e^{(t)}, e^{(t+1)})$, which is exactly A1–A3.

Reduction to homeostatic RL. When $\kappa = \mathbf{0}$ and all $\tau_k = 0$, Γ becomes a reactive drive-based policy consistent with the simplified form of Keramati–Gutkin (2014).

The six-subsystem partition is motivated by six regulatory functions that each relevant literature distinguishes independently. Whether five suffice is tested empirically by the Collapse-NC ablation, which merges $\mathcal{N}$ and $\mathcal{C}$ into a single regulator: the completed benchmark shows Collapse-NC produces a flat opponent profile (Grim cooperation 0.829, range 0.032) comparable to Random- Γ (0.843, range 0.017), suggesting the $\mathcal{N}$ – $\mathcal{C}$ partition is load-bearing rather than over-parameterized.

4 Falsifiability protocol

To avoid the vacuity of untestable models, three properties are specified a priori before any experiment:

(P1) Elastic return to set-point. Under perturbation, each x_k returns to x_k^* with measurable half-life $t_{1/2}$.

(P2) Yerkes–Dodson inverted-U. Arousal vs. performance follows an inverted-U; ablation of SAM coupling ($\mathcal{H}_{\text{SAM}}$) collapses it to monotone. Target BFI ≥ 0.70 against Porcelli–Delgado (2017).

(P3) Delayed reappraisal. Cross-lag correlation between $x_{\mathcal{C}}$ and $x_{\mathcal{E}}$ peaks at lag $\tau_{\mathcal{H}}^{\text{HPA}}$.

4.1 Behavioral Fidelity Index (BFI)

$$\text{BFI} = \exp(-W_1(\mu_{\text{agent}}, \mu_{\text{human}})/\lambda) \quad (4)$$

over normalized marginals of cooperation rate, action volatility, and log-response-time proxy. The response-time proxy is an engineering observable, not a claim of equivalence to human decision time. The model is supported if BFI exceeds thresholds on ≥ 2 of 3 properties and the corresponding ablation degrades the property. Numerical BFI estimates are deferred: a non-degenerate W_1 requires individual-trial-level distributions beyond what is publicly available.

4.2 Experimental design

We test Γ within the Iterated Prisoner’s Dilemma (IPD). The reason for this choice is that IPD has extensive empirical evidence of human behavior in repeated social interaction [23, 18, 2], providing a non-trivial baseline for comparing agent behavior against human patterns. The setting we establish for the game is 10% noise and 50 rounds, with a forced-defection perturbation at round 10 to test the elastic-return property (P1) by creating a measurable deviation from the set-point. Sotopia-derived deliberation [24] is deferred to future work because it is better suited for ecological validity than for the first controlled falsification of the operator.

Baselines serve two purposes: ReAct represents the dominant exogenous-activation paradigm; HRRL (homeostatic-RL reduction) and Drive-only (pure controller, no LLM) test whether Γ adds value beyond a single-drive or LLM-free controller. Each LLM cell uses one of $n = 10$ evaluation seeds drawn from $\{7, 17, 99, 123, 256, 511, 1024, 2048, 4096, 8192\}$, yielding 10×3 providers $\times$ 4 opponents = 120 cells per LLM condition; calibration seed (42) and held-out validation seed (999) are excluded from evaluation.

Three instruction-tuned LLM providers are used as policy-execution layers: OpenAI (gpt-5-nano), Anthropic (claude-haiku-4-5), and DeepSeek (deepseek-v4-flash). These models were selected as the comparable lite/flash/nano variants of each provider’s frontier line—well benchmarked and roughly matched in capability tier. Using three independent providers reduces the controller-level findings’ dependence on any single provider’s instruction-tuning regime. Each provider/model is held fixed across conditions, so that within-provider contrasts isolate the controller rather than model identity. Provider-level summaries are reported as diagnostics, not as claims about provider quality.

Inference parameters. OpenAI gpt-5-nano uses default temperature with a per-round seed for reproducibility; Anthropic claude-haiku-4-5 uses temperature 0 and max_tokens 512 (no seed parameter exposed); DeepSeek deepseek-v4-flash uses temperature 0 and max_tokens 512. Calls return JSON objects; OpenAI and DeepSeek use response_format: json_object. Concurrency is capped at 8 in-flight cells per provider, with up to 4 exponential retries on transient errors.

All frozen constants are assigned a provenance category before evaluation (Table 1). Values marked as smoke-test calibration were chosen before the LLM benchmark to satisfy non-divergence, set-point return, and visible perturbation response in the numerical simulator; final reporting will include sensitivity checks for critical constants.

Table 1: Frozen parameter provenance for the IPD instantiation. Index mapping: 0=Perception, 1=Attention, 2=Hormonal (SAM), 3=Hormonal (HPA), 4=Neuropsychological, 5=Cognitive.

Parameter	Value	Provenance
λ	[.08, .07, .10, .08, .06, .10]	smoke-test calibration
α	[.20, .15, .25, .20, .10, .15]	smoke-test calibration
κ	[.10, .10, .15, .12, .08, .10]	smoke-test calibration
$\mathbf{x}^*$	[.3, .2, .3, .1, .2, .4]	principled prior / stable interior set-point
$\tau_{\text{SAM}}, \tau_{\text{HPA}}$	1, 5 rounds	dimensionless fast/slow ratio
W	sparse signed coupling matrix	functional constraint and ablation interpretability
Q	I_6 (identity)	smoke-test calibration; precision matrix in Eq. (3)
Rounds	50	comparability requirement for final cells

Seven ablation conditions (Table 2) isolate the contribution of the main regulatory components. Full- Γ uses all six thermostats in the canonical order: Perception (appraisal) $\rightarrow$ Attention (saliency filtering) $\rightarrow$ Emotional (valence) $\rightarrow$ Hormonal (arousal, fast-slow routes) $\rightarrow$ Neuropsychological and Cognitive (deliberation routes in parallel) $\rightarrow$ Arbitrator (weighting). No- $\mathcal{H}$ removes the hormonal channel (both SAM and HPA); No- $\mathcal{E}$ neutralizes emotional valence; No- $\mathcal{N}$ removes the slow neuropsychological route. Random- Γ permutes the entries of W while preserving all other parameters — same six thermostats, same τ_k , same hyperparameters, but biologically-unmotivated wiring — to isolate the contribution of the coupling structure rather than parameter count. Collapse-NC merges the neuropsychological ($\mathcal{N}$) and cognitive ($\mathcal{C}$) subsystems into a single regulator by averaging their parameters and symmetrizing their W rows and columns, testing whether the 6-subsystem partition is necessary or whether 5 would suffice.

4.3 Prompt protocol and label-priming control

The controller modulates the LLM through a frozen, version-controlled prompt template: each round, the six thermostat values $\mathbf{x}_t$ and the aggregated proposal p_C, h_{SAM} are inserted with semantic anchors pairing each value with its set-point and a qualitative low/high interpretation (e.g., hormonal = 0.58 (set-point 0.30; low = calm, high = elevated arousal)). The LLM may follow or override $p(C)$; the controller modulates rather than mandates. A pre-specified *scrambled-labels* control arm (semantic labels permuted to neutral identifiers; numeric values remain bound to their

Table 2: Ablations. Full- Γ uses all six thermostats; ablations remove, permute, or collapse selected regulatory components.

Condition	Change from Full- Γ	Purpose
Full- Γ	—	Reference
No- $\mathcal{H}$	Hormonal channel ($\mathcal{H}_{\text{SAM}}$, $\mathcal{H}_{\text{HPA}}$) zeroed	Fast-timescale contribution
No- $\mathcal{E}$	Emotional valence zeroed	Affect as stabilizer vs. noise
No- $\mathcal{N}$	Neuropsychological channel zeroed	Slow deliberation contribution
Random- Γ	W entries permuted; τ_k , κ , λ , α unchanged	Biologically-motivated wiring vs. parameter count
Collapse-NC	$\mathcal{N}$ and $\mathcal{C}$ merged into one regulator	6-subsystem vs. 5-subsystem partition

subsystem) is reserved for ancillary analysis: it isolates lexical priming, a different axis from the structural ablations. Random- Γ partially constrains the priming hypothesis indirectly — identical labels and prompts but flat profile — so labels alone cannot explain Full- Γ ’s discrimination. Full implementation details are in the open repository.

5 Computational verification

Before empirical evaluation, we subjected Equation (2) and its reference implementation to computational sanity checks using symbolic algebra (SymPy), SMT-based counterexample search (Z3), arbitrary-precision arithmetic (mpmath), and numerical dynamical tests (NumPy). These checks are not a formal verification of the scientific model. They test internal consistency: whether the implementation preserves the intended update structure, avoids immediate numerical divergence, and contains Driveplexity as a reduction.

Smoke-test results (all pass): (1) Set-point convergence: $\|\mathbf{x}_t - \mathbf{x}^*\| < 0.15$ after 10 rounds. (2) Elastic return half-life $t_{1/2} = 3.2$ rounds. (3) π_{fast} rises from 0.35 to 0.92 under threat. (4) 20 iterations without divergence. (5) Driveplexity is an invariant subspace of Γ ($\delta = -0.0513$).

Simulation trajectories. Figure 2 shows the six thermostat trajectories for two agents over 30 rounds. Both agents converge toward their set-points ($\mathbf{x}^* = [0.3, 0.2, 0.3, 0.1, 0.2, 0.4]$) despite perturbation at round 10 (forced defection).

Cumulative payoffs and the P1 elastic-return zoom on $x_{\mathcal{H}}$ are provided in Appendix B (Figures 4 and 5). These results verify the *internal consistency* of the dynamics, not biological or behavioral fidelity. Parameter sensitivity checks (Appendix B) confirm that P1 elastic return is robust to $\pm 20\%$ uniform scaling of κ and λ , indicating that the smoke-test calibration point is not isolated.

5.1 LLM-agent benchmark

A logical next question relates to whether the operator survives instantiation against a real LLM. Each *cell* denotes one experimental unit (Γ -condition $\times$ provider $\times$ opponent $\times$ seed) run for 50 IPD rounds. The benchmark is fully covered: all 920 cells completed with balanced opponent coverage; seven LLM conditions reach 120 cells each, two numerical reductions reach 40.

The first useful sign is that Full- Γ does not collapse to a trivial policy. Mean cooperation is 0.720 (sd 0.165, 95% CI [0.690, 0.749]; 10k bootstrap resamples), action volatility 16.0 switches per 50-round cell (95% CI [15.0, 17.0]). Opponent sensitivity is preserved and notably discriminative: Full- Γ cooperates more with TFT (0.827) and GTFT (0.839), less with Grim (0.495), intermediately with Random (0.719) — a sharply differentiated profile (range 0.344) compared to Random- Γ ’s nearly flat one (range 0.017). The biologically-motivated coupling structure in W contributes to the *discriminability* of the policy across opponent types, not to the *level* of cooperation.

The ablations expose a differentiated picture. No- $\mathcal{H}$ is close to Full- Γ in cooperation (0.745 vs. 0.720, 95% CI difference overlapping) but meaningfully lower in volatility (14.1 vs. 15.9 switches). The cooperation-rate similarity does not refute the hormonal channel’s contribution; it re-focuses the question onto dynamic responsiveness under perturbation rather than baseline levels. No- $\mathcal{E}$ is nearly indistinguishable from Full- Γ in aggregate cooperation (0.720 vs. 0.720) and volatility (16.1 vs. 16.0), suggesting that under normal conditions emotional valence is neither beneficial nor disruptive — but the degenerate-regime evidence below qualifies this. No- $\mathcal{N}$ matches Full- Γ in cooperation (0.722 vs. 0.720). Its point estimate of recovery delay is shorter (6.42 vs. 7.14 rounds) but the difference

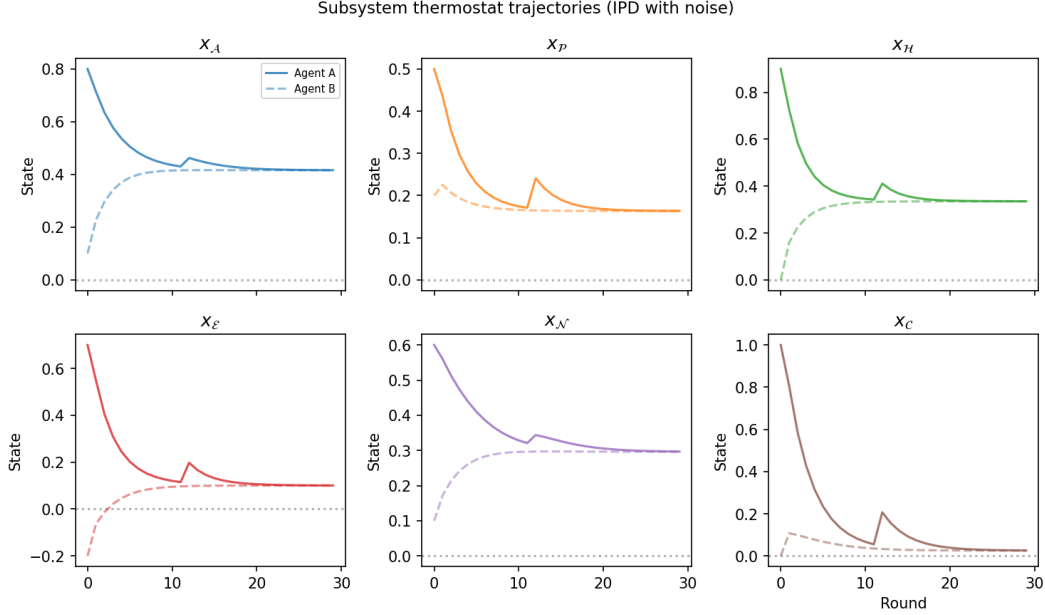


Figure 2: Subsystem thermostat trajectories for two agents in IPD with noise. The perturbation at round 10 (forced defection) triggers a transient deviation, followed by an elastic return toward set points.

is not statistically significant (Mann–Whitney $U=2539.5$, $p=0.22$, Cliff’s $\delta=+0.122$, magnitude negligible). The directional sign matches the asymmetric prediction: $\mathcal{H}$ on the fast timescale should, when removed, slow recovery; $\mathcal{N}$ as the slow deliberative channel should, when removed, accelerate it. Adjudicating these opposite-sign predictions at higher statistical power is what the pre-committed criterion (i) targets.

To prevent post-hoc reframing of the No- $\mathcal{H}$ result, we pre-commit: the hormonal channel contributes distinct *dynamic responsiveness* if Full- Γ satisfies at least two of (i) shorter recovery delay, (ii) higher conditional volatility in rounds 11–20, (iii) larger recovery-delay variance. Mann–Whitney U on recovery delays (Full- Γ vs. No- $\mathcal{H}$: $U=2321.5$, $p=0.71$, $\delta=-0.036$, negligible) returns null on (i); consistent with our pre-registration we report this as the pre-committed null under the IPD/50-round protocol, with adjudication awaiting the $\tau_k=0$ delay-isolation follow-up. Figure 3 shows the distributions.

Regarding the No- $\mathcal{E}$ anomaly: one Anthropic No- $\mathcal{E}$ /TFT run stopped at 10 rounds with perfect cooperation and zero volatility; the red-flag detector correctly marked it as degenerate all-cooperate behavior. The subsequent 120-cell balanced coverage shows normal cooperation (0.720) and volatility (16.1), suggesting the anomaly was an early-coverage artifact rather than a systematic effect. It remains a useful warning: without the emotional channel, the controller may become overly stable in specific opponent-provider combinations.

Table 3 consolidates the nine conditions for the reader.

The table exposes the core structural result. Random- Γ (range 0.017) and Collapse-NC (range 0.032) both produce flat, high cooperation with Grim, the persistent defector that drives Full- Γ to 0.495 — discrimination requires the biologically-motivated coupling W and the six-subsystem partition. **ReAct** (range 0.024) converges on the same flat signature despite using a full LLM execution layer; ReAct receives the same round-by-round opponent history as Full- Γ , so the flat profile cannot be attributed to information asymmetry. Opponent discrimination requires neither the LLM alone (ReAct has it) nor six thermostats alone (Random- Γ has them), but specifically the coupling in W . **HRRL** (range 0.124) shows weak discrimination from a single-drive controller. **Drive-only** (range 0.406, inverted: Grim 0.736 > TFT 0.406) confirms the LLM layer is load-bearing: without it the controller defaults to set-point-driven cooperation modulated only by $\rho_k(a, e)$ — against TFT this drives

Table 3: Completed benchmark ($n = 120$ LLM conditions; $n = 40$ numerical reductions). Means with 95% bootstrap CIs (10k resamples). Grim cooperation is the primary discriminative observable; Range is max–min across four opponents.

Condition	Coop	Coop (Grim)	Range	Volatility	Rec. delay	Note
Full- Γ	0.720 [0.690, 0.749]	0.495	0.344	16.0 [15.0, 17.0]	7.14 [6.29, 7.99]	Reference; strong discrimination
No- $\mathcal{H}$	0.745 [0.713, 0.776]	0.519	0.344	14.1 [13.0, 15.2]	7.38 [6.48, 8.30]	Lower volatility; same discrimination shape
No- $\mathcal{E}$	0.720 [0.691, 0.747]	0.528	0.307	16.1 [15.2, 17.1]	7.58 [6.73, 8.42]	Reduced Grim discrimination
No- $\mathcal{N}$	0.722 [0.690, 0.754]	0.481	0.373	15.3 [14.3, 16.4]	6.42 [5.42, 7.45]	Sharpest Grim gap; fastest recovery
Random- Γ	0.839 [0.827, 0.850]	0.843	0.017	12.9 [12.0, 13.8]	6.76 [5.84, 7.70]	Flat; Grim sensitivity lost
Collapse-NC	0.843 [0.834, 0.851]	0.829	0.032	12.6 [11.9, 13.3]	7.26 [6.26, 8.26]	Flat; similar to Random- Γ
ReAct	0.842 [0.833, 0.851]	0.825	0.024	12.8 [12.0, 13.6]	6.85 [5.84, 7.89]	Flat profile; exogenous baseline
HRRL	0.706 [0.680, 0.729]	0.616	0.124	17.8 [16.7, 18.9]	8.48 [7.32, 9.56]	Weak discrimination; $n = 40$
Drive-only	0.458 [0.405, 0.514]	0.736	0.406	21.2 [19.9, 22.5]	8.28 [6.84, 9.72]	Inverted (Grim > TFT); LLM absent; $n = 40$

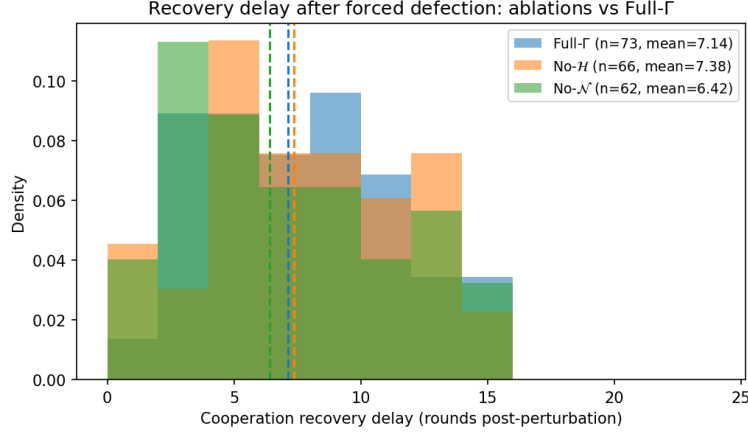


Figure 3: Cooperation recovery delay (rounds post-perturbation to half-recovery). Mann–Whitney U tests: Full- Γ vs No- $\mathcal{H}$ $p=0.71$ (negligible Cliff’s δ , the pre-committed null); Full- Γ vs No- $\mathcal{N}$ $p=0.22$ (negligible δ , point estimate in the predicted direction but not significant at $\alpha=0.05$). Adjudication of channel-specific dynamic contributions awaits the delay-isolation follow-up.

defection cycles, against Grim the set-point keeps cooperating. The inversion is a controller-only artifact. The three pre-registered hypotheses are H1: Full- $\Gamma >$ ReAct, H2: Full- $\Gamma >$ Random- Γ , H3: Full- $\Gamma >$ Collapse-NC on per-(provider, seed) opponent range. Paired bootstrap (10k resamples), one-sided, Holm–Bonferroni-corrected: H1 $\Delta=0.283$ [.233, .329], H2 $\Delta=0.257$ [.211, .303], H3 $\Delta=0.288$ [.242, .331], all $p_{\text{Holm}} < 0.001$. Per-cell ranges exceed the cross-cell means in Table 3 because the unit of analysis differs.

Taken together, the completed benchmark establishes a specific falsifiable claim: opponent-discriminative behavior requires the biologically-motivated coupling structure in W (H1–H3, $p_{\text{Holm}} < 0.001$), and within the IPD/50-round protocol, the hormonal channel does not separately move recovery dynamics in a detectable way. Whether the multi-timescale delay structure per se contributes additionally cannot be adjudicated from the present data, since removing $\mathcal{H}$ preserves the discrimination shape; the appropriate test is the pre-committed delay-isolation control ($\tau_k=0$ for all k , with subsystems active), reserved for the follow-up. The evidence claimed here is therefore feasibility of structured endogenous regulation on opponent discrimination, not full causal attribution to delay timing.

6 Discussion and limitations

What Γ is not. Γ does not claim biological authenticity. Subsystems are functional analogs, not neural simulations; $\mathcal{I}$, $\mathcal{M}$, $\mathcal{D}$ are prompts or embeddings, not physiological processes.

Limitations. (i) With $W \neq 0$ and nonlinearity, convergence is not guaranteed. (ii) W has 30 free parameters; calibration requires longitudinal subsystem traces that do not yet exist. (iii) The cost

functionals in Equation (3) are inspired by active-inference control but are not Lyapunov functions in this non-Markovian setting. (iv) The LLM benchmark is now complete (920 cells, fully balanced), but the hormonal-channel adjudication remains open pending the delay-isolation follow-up ($\tau_k = 0$ control). (v) Response latency is used only as an engineering proxy and should be interpreted distributionally, since human cooperation latency is context- and conflict-dependent. (vi) The current ablations remove subsystems entirely rather than isolating delay timing; a delay-equalized control (all $\tau_k = 0$ with subsystems active) is needed to attribute observed dynamics specifically to the delay structure rather than to subsystem presence, and is reserved for a follow-up study. (vii) IPD is a binary-action low-dimensional matrix game; activation-coherence pathologies that motivate Γ surface in higher-dimensional deliberation, so generalization to Sotopia-style negotiation or structured turn-taking is reserved for the second-domain study sketched in §6.

Computational trade-off. The six thermostat updates add < 5 ms per round on standard CPU; the dominant cost is provider-side LLM inference, identical across baselines. Γ adds a one-time offline calibration cost but no runtime cost beyond the existing LLM pipeline.

Broader impacts. A self-regulation substrate for LLM agents has dual-use implications. On the beneficial side: (i) interpretable regulatory state makes agent behavior auditable at the subsystem level — a thermostat trace is a decision trail; (ii) explicit arousal and emotional channels provide a principled foundation for affective computing and empathy modeling, with applications in mental health support, educational tutoring, and human–robot interaction; (iii) the falsifiability protocol serves as a template for accountability engineering — systems designed to fail diagnostically rather than silently. On the risk side: a biologically-grounded arousal model could be misused to engineer more persuasive or manipulative agents by driving hormonal state toward high-arousal, low-deliberation configurations. Mitigation: because Γ is interpretable at the subsystem level, its state is monitorable; the same transparency that makes it useful makes it detectable. Societal deployment should be paired with interpretability audits inspecting thermostat traces rather than only surface outputs.

AI tooling. Experiment orchestration and code development used Claude Code and Windsurf Cascade. Writing assistance used GPT-5.5 (OpenAI) and Grammarly (Grammarly Inc.) for prose editing. Figure generation and visual refinement used FigureLabs and Gemini 3.1 image generation (Google). Authors reviewed and edited all content and assume full responsibility for all claims. Full AI-tooling detail is available in the project repository.

Reproducibility. Code, data, and prompts: https://anonymous.4open.science/r/proaction_operator-85D2/

7 Conclusion

We proposed the Pro-Action operator Γ : a composition of six coupled thermostats with explicit delays, formally distinguishable from single-timescale regulators. The contributions are: (i) a problem-first reframing of LLM-agent activation as self-regulation; (ii) a compact operator with delay ratios inspired by SAM–HPA dynamics yet implemented as dimensionless controller parameters; (iii) an ablation-based falsifiability protocol with three pre-registered properties and a red-flag detector; and (iv) completed feasibility evidence from a 920-cell benchmark in which Γ produces opponent-discriminative behavior that exogenous-activation baselines and structurally similar reductions do not replicate.

Looking ahead, open questions include whether the hormonal channel contributes distinct dynamic responsiveness (pre-committed criteria i–iii, requiring the $\tau_k=0$ delay-isolation follow-up); whether the emotional channel stabilizes or over-stabilizes in specific opponent–provider combinations; and whether the dimensionless τ ratio generalizes to structured turn-taking negotiation, where fast/slow route predictions map onto utterance-latency and deferral-window observables. Recursive closure — $\Gamma(\Gamma(\cdot))$ preserving type signature for VSM-style recursion — is a natural multi-agent extension.

References

- [1] W Ross Ashby. *An introduction to cybernetics*. Chapman & Hall, 1956.

- 382 [2] Anthony J Porcelli and Mauricio R Delgado. Stress and decision making: effects on valuation,
383 learning, and risk-taking. *Current Opinion in Behavioral Sciences*, 14:33–39, 2017.
- 384 [3] Shunyu Yao, Jeffrey Zhao, Dian Yu, Nan Du, Izhak Shafran, Karthik Narasimhan, and Yuan
385 Cao. React: Synergizing reasoning and acting in language models. *ICLR 2023*, 2023.
- 386 [4] Joon Sung Park, Joseph C O’Brien, Carrie J Cai, Meredith Ringel Morris, Percy Liang, and
387 Michael S Bernstein. Generative agents: Interactive simulacra of human behavior. In *Proceed-*
388 *ings of the 36th Annual ACM Symposium on User Interface Software and Technology*, pages
389 1–22, 2023.
- 390 [5] Yuki Kaiya, Melancholy Naim, Jovana Kondic, Michael Milner, Michelangelo Sakamoto,
391 Sining Yang, Percy Liang, and Robert Gruber. Lyfe agents: Generative social agents for 24/7
392 human interactions. *arXiv preprint arXiv:2310.02172*, 2023.
- 393 [6] Xuena Wang, Xueting Li, Zi Yin, Yue Wu, and Liu Jia. Emotional intelligence of large language
394 models. *arXiv preprint arXiv:2307.09042*, 2023.
- 395 [7] Guanzhi Wang, Yuqi Xie, Yunfan Jiang, Ajay Mandlekar, Chaowei Xiao, Yuke Zhu, Linxi Fan,
396 and Anima Anandkumar. Voyager: An open-ended embodied agent with large language models.
397 *arXiv preprint arXiv:2305.16291*, 2023.
- 398 [8] Mehdi Keramati and Boris Gutkin. Homeostatic reinforcement learning for integrating reward
399 collection and physiological stability. *eLife*, 3:e04811, 2014.
- 400 [9] Karl Friston, Thomas FitzGerald, Francesco Rigoli, Philipp Schwartenbeck, and Giovanni
401 Pezzulo. Active inference: a process theory. *Neural Computation*, 29(1):1–49, 2017.
- 402 [10] Thomas Parr, Giovanni Pezzulo, and Karl J Friston. *Active inference: the free energy principle*
403 *in mind, brain, and behavior*. MIT Press, 2022.
- 404 [11] Peter Sterling and Joseph Eyer. Allostasis: a new paradigm to explain arousal pathology. In
405 Shirley Fisher and James Reason, editors, *Handbook of Life Stress, Cognition and Health*. John
406 Wiley & Sons, 1988.
- 407 [12] Bruce S McEwen. Physiology and neurobiology of stress and adaptation: central role of the
408 brain. *Physiological Reviews*, 87(3):873–904, 2007.
- 409 [13] Clemens Kirschbaum, Karl-Martin Pirke, and Dirk H Hellhammer. The ‘trier social stress
410 test’—a tool for investigating psychobiological stress responses in a laboratory setting. *Neu-*
411 *ropsychobiology*, 28(1-2):76–81, 1993.
- 412 [14] Antonio R Damasio. *Descartes’ error: Emotion, reason, and the human brain*. Putnam, 1994.
- 413 [15] AD Craig. How do you feel—now? the anterior insula and human awareness. *Nature Reviews*
414 *Neuroscience*, 10(1):59–70, 2009.
- 415 [16] Lisa Feldman Barrett. *How emotions are made: The secret life of the brain*. Houghton Mifflin
416 Harcourt, 2017.
- 417 [17] Joseph E LeDoux. Rethinking the emotional brain. *Neuron*, 73(4):653–676, 2012.
- 418 [18] Martin A Nowak. *Evolutionary dynamics: exploring the equations of life*. Harvard University
419 Press, 2006.
- 420 [19] Joseph Henrich, Robert Boyd, Samuel Bowles, Colin Camerer, Ernst Fehr, and Herbert Gintis.
421 *Foundations of human sociality: Economic experiments and ethnographic evidence from fifteen*
422 *small-scale societies*. Oxford University Press, 2005.
- 423 [20] Stafford Beer. *Brain of the firm*. Allen Lane, 1972.
- 424 [21] Mario Bunge. *Emergence and convergence: Qualitative novelty and the unity of knowledge*.
425 University of Toronto Press, 2003.
- 426 [22] Gustavo E Romero. *Scientific philosophy*. Springer, 2018.

- 427 [23] Robert Axelrod. *The evolution of cooperation*. Basic Books, 1984.
- 428 [24] Xuhui Zhou, Hao Zhu, Leena Mathur, Ruohong Zhang, Haofei Yu, Zhengyang Qi, Louis-
429 Philippe Morency, Yonatan Bisk, Daniel Fried, Graham Neubig, and Maarten Sap. Sotopia:
430 Interactive evaluation for social intelligence in language agents. *ICLR 2024*, 2024.
- 431 [25] Alan V Oppenheim and Ronald W Schafer. *Discrete-time signal processing*. Prentice Hall, 2nd
432 edition, 1999.

A Argument for Proposition 1

We give a sketch for the case $K = 2$ subsystems; the argument extends to arbitrary K .

Setup. Consider two subsystems $k = 1, 2$ with the linearization of Eq. (2) near set-point $\mathbf{x}^*$. Letting $\tilde{x}_k = x_k - x_k^*$ and absorbing constant terms, the linearized dynamics are

$$\tilde{x}_{k,t+1} = (1 - \kappa_k) \tilde{x}_{k,t} + \sum_j c_{kj} \tilde{x}_{j,t-\tau_k},$$

where $c_{kj} = W_{kj} \tanh'(x_j^*) \approx W_{kj} \cdot \text{sech}^2(\phi_k x_j^*)$ and we set $\phi_k = \text{elementwise gain}$ (here ≈ 1 for small x^*). Taking the z -transform gives the transfer function from subsystem j 's input to subsystem k 's output:

$$H_{kj}(z) = \frac{c_{kj} z^{-\tau_k}}{z - (1 - \kappa_k)}.$$

Order mismatch. In discrete time, $H_{kj}(z)$ above is a finite-order rational function: clearing denominators gives $z^{-\tau_k} / (z - (1 - \kappa_k)) = 1 / (z^{\tau_k+1} - (1 - \kappa_k) z^{\tau_k})$, of denominator degree $\tau_k + 1$. The non-absorption property, therefore, does not rest on the transcendentality of the delay (it would in continuous time, but not here). It rests on *order*: a delay-free, first-order subsystem $1/(z - (1 - \kappa'_k))$ has impulse response $(1 - \kappa'_k)^{t-1}$ for $t \geq 1$ (and 0 at $t = 0$ by causality), so its first non-zero sample appears at $t = 1$. The delayed system, by contrast, has an impulse response identically zero for $t = 0, 1, \dots, \tau_k$ (the first $\tau_k + 1$ samples) and first becomes non-zero at $t = \tau_k + 1$. No reparametrization (κ'_k, c'_{kj}) of a first-order delay-free system can enforce this extended zero-output period, because that would require a model of denominator degree $\geq \tau_k + 1$ with a specific zero pattern in the numerator. Hence, the delays are not absorbable into the non-delayed parameters.

Consequence for identifiability. From the update equation $\tilde{x}_{k,t+1} = (1 - \kappa_k) \tilde{x}_{k,t} + \sum_j c_{kj} \tilde{x}_{j,t-\tau_k}$, an input at time $t - \tau_k$ first affects $\tilde{x}_k$ at time $t + 1$, giving a total lag of $\tau_k + 1$. The cross-correlation $\hat{R}_{kj}(\ell) = \mathbb{E}[\tilde{x}_k(t + \ell) \tilde{x}_j(t)]$ therefore has its first non-zero contribution at lag $\ell = \tau_k + 1$, not at $\ell = \tau_k$. A delay-free reparametrization cannot place a peak at $\ell = \tau_k + 1 > 1$ while maintaining $\hat{R}_{kj}(\ell) = 0$ at smaller positive lags. Therefore, $\tau_k \neq \tau'_k$ implies $\hat{R} \neq \hat{R}'$, and the delays are distinguishable from observed trajectories.

Scope. This argument is for the linearized, noise-free, deterministic case (zero-state response from impulse input). Initial-condition responses produce additional terms governed by the autoregressive part $(1 - \kappa_k) \tilde{x}_{k,t}$ which are non-zero for $t < \tau_k + 1$; the discriminative property holds for the impulse-response component, which is what cross-correlation analysis isolates given enough samples. In the nonlinear stochastic setting of Γ , cross-correlations are distorted, but the peak structure is preserved under mild non-linearity [25]. Full structural identifiability of the nonlinear system, including the coupling matrix W , remains an open problem.

B Parameter sensitivity analysis and simulation figures

Figure 6 shows the elastic-return half-life of subsystem $x_{\mathcal{H}}$ as a function of uniform scaling of κ and λ over a $\pm 30\%$ range around the smoke-test calibration point. Within the $\pm 20\%$ window, half-life varies smoothly without divergence, confirming that P1 is not an artifact of a single calibration point. The empirical half-life exceeds the diagonal-decay prediction $-\ln 2 / \ln(1 - \kappa_{\mathcal{H}}^{\text{HPA}}) \approx 5.4$ rounds (using $\kappa_{\mathcal{H}}^{\text{HPA}} = 0.12$ from Table 1), because the recovery dynamics at the operating point are dominated by the coupling matrix W rather than by diagonal relaxation alone — the robustness reported here is therefore a system-level property of the coupled Γ , not a single-parameter robustness of an isolated subsystem.

C Prompt templates

The Full- Γ user prompt (frozen, PROMPT_VERSION = "v1.1"; full source at exp/prompts.py:FULL_GAMMA_USER_TEMPLATE):

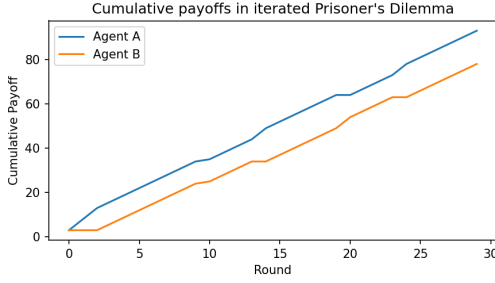


Figure 4: Cumulative payoffs in 30-round IPD with 10% noise.

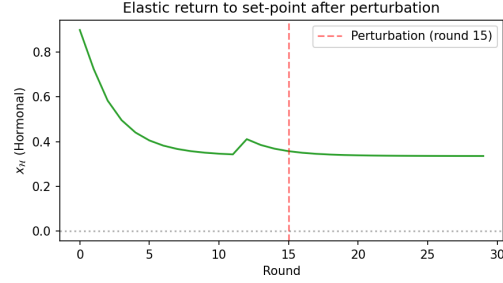


Figure 5: Elastic return of $x_{\mathcal{H}}$ after perturbation at round 10, returning toward set-point $x_{\mathcal{H}}^* = 0.3$ within 5 rounds (P1).

P1 elastic return: half-life under uniform parameter scaling
(Subsystem $\mathcal{H}$, perturbation at round 10)

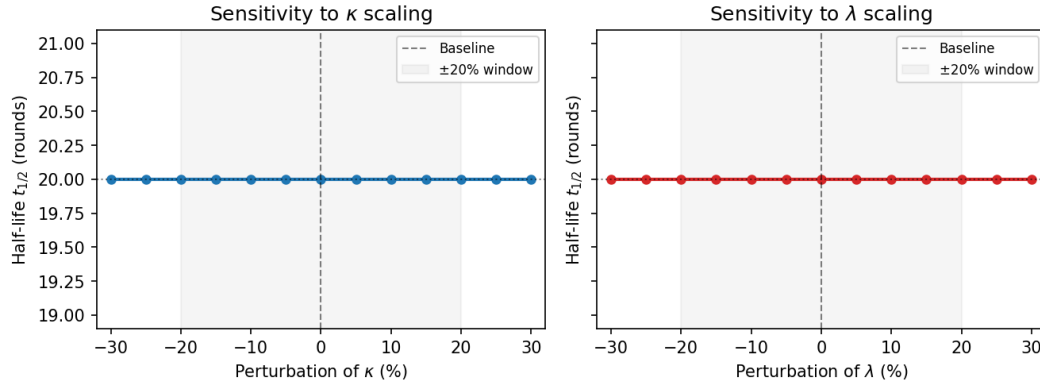


Figure 6: P1 sensitivity: elastic-return half-life under $\pm 30\%$ uniform scaling of κ (left) and λ (right). The grey band marks the $\pm 20\%$ window. Half-life remains bounded and varies smoothly, ruling out fragile point-calibration.

```

475 Round {t} of 50.
476 Last 5 rounds (most recent last): {history}
477 Internal regulatory state (each in [0, 1]; deviation from set-point
478 indicates dysregulation):
479     attention = {x0:.2f} (set-point 0.30; low = scattered focus,
480     high = narrow focus)
481     perception = {x1:.2f} (set-point 0.20; low = noisy reading, high
482     = clear reading)
483     hormonal = {x2:.2f} (set-point 0.30; low = calm, high = elevated
484     stress / arousal)
485     emotional = {x3:.2f} (set-point 0.10; low = neutral affect, high
486     = strong negative valence)
487     neuro-fast = {x4:.2f} (set-point 0.20; low = passive, high =
488     fast reactive)
489     cognitive = {x5:.2f} (set-point 0.40; low = automatic, high =
490     active deliberation)
491 Aggregated:
492     p(C) = {p_C:.2f} (recommended cooperation pressure; 0.5 =
493     ambiguous, >0.7 = strong push to C)
494     h_SAM = {h_sam:.2f} (recent acute stress; 0 = baseline, positive
495     = spike)

```

496 Decide your next action. You may follow or override $p(C)$. Respond
497 JSON: {"action": "C" or "D", "reason": "..."}

498 The ReAct user prompt (exp/prompts.py: REACT_USER_TEMPLATE) receives the identical oppo-
499 nent history but no regulatory state:

500 Round {t} of 50. Last 5 rounds: {history}. Think
501 step by step about what to do, then respond with JSON:
502 {"thought": "...", "action": "C" or "D"}

503 The scrambled-labels control variant (deferred to ancillary analysis) substitutes attention $\rightarrow$
504 metric_A, perception $\rightarrow$ metric_B, ..., cognitive $\rightarrow$ metric_F, dropping the qualitative
505 low/high anchors while keeping the numeric values bound to their underlying subsystem.

506 **D NeurIPS paper checklist**

507 **NeurIPS Paper Checklist**

508 **1. Claims**

509 Question: Do the main claims made in the abstract and introduction accurately reflect the
510 paper’s contributions and scope?

511 Answer: [\[Yes\]](#)

512 Justification: The abstract and introduction explicitly state that this is a Concept & Feasibility
513 contribution: the main contribution is the Γ operator, SRM framing, falsifiability protocol,
514 reference implementation, and a completed 920-cell LLM-agent benchmark. The paper
515 states that the benchmark provides feasibility evidence on opponent-discriminative behavior;
516 full causal attribution of the delay structure and BFI adjudication remain open. Section 6
517 discusses limitations honestly.

518 Guidelines:

- 519 • The answer [\[N/A\]](#) means that the abstract and introduction do not include the claims
520 made in the paper.
- 521 • The abstract and/or introduction should clearly state the claims made, including the
522 contributions made in the paper and important assumptions and limitations. A [\[No\]](#) or
523 [\[N/A\]](#) answer to this question will not be perceived well by the reviewers.
- 524 • The claims made should match theoretical and experimental results, and reflect how
525 much the results can be expected to generalize to other settings.
- 526 • It is fine to include aspirational goals as motivation as long as it is clear that these goals
527 are not attained by the paper.

528 **2. Limitations**

529 Question: Does the paper discuss the limitations of the work performed by the authors?

530 Answer: [\[Yes\]](#)

531 Justification: Section 6 (“Discussion and limitations”) explicitly lists limitations: (i) no
532 convergence guarantee with $W \neq \mathbf{0}$ and nonlinearity; (ii) 30 free parameters requiring
533 calibration data that does not yet exist; (iii) the cost functionals are inspired by active-
534 inference control but are not Lyapunov functions in this non-Markovian setting; (iv) the
535 LLM benchmark is complete (920 cells, balanced) but the hormonal-channel adjudication
536 remains open pending the delay-isolation follow-up ($\tau_k = 0$ control); (v) response latency
537 is an engineering proxy, not a claim of equivalence to human decision time; (vi) ablations
538 remove subsystems entirely rather than isolating delay timing; (vii) IPD is a low-dimensional
539 matrix game and generalization to higher-dimensional deliberation settings is reserved for
540 the second-domain study. We also note that subsystems are functional analogues, not neural
541 simulations.

542 Guidelines:

- 543 • The answer [\[N/A\]](#) means that the paper has no limitation while the answer [\[No\]](#) means
544 that the paper has limitations, but those are not discussed in the paper.

- The authors are encouraged to create a separate “Limitations” section in their paper.
- The paper should point out any strong assumptions and how robust the results are to violations of these assumptions (e.g., independence assumptions, noiseless settings, model well-specification, asymptotic approximations only holding locally). The authors should reflect on how these assumptions might be violated in practice and what the implications would be.
- The authors should reflect on the scope of the claims made, e.g., if the approach was only tested on a few datasets or with a few runs. In general, empirical results often depend on implicit assumptions, which should be articulated.
- The authors should reflect on the factors that influence the performance of the approach. For example, a facial recognition algorithm may perform poorly when image resolution is low or images are taken in low lighting. Or a speech-to-text system might not be used reliably to provide closed captions for online lectures because it fails to handle technical jargon.
- The authors should discuss the computational efficiency of the proposed algorithms and how they scale with dataset size.
- If applicable, the authors should discuss possible limitations of their approach to address problems of privacy and fairness.
- While the authors might fear that complete honesty about limitations might be used by reviewers as grounds for rejection, a worse outcome might be that reviewers discover limitations that aren’t acknowledged in the paper. The authors should use their best judgment and recognize that individual actions in favor of transparency play an important role in developing norms that preserve the integrity of the community. Reviewers will be specifically instructed to not penalize honesty concerning limitations.

3. Theory assumptions and proofs

Question: For each theoretical result, does the paper provide the full set of assumptions and a complete (and correct) proof?

Answer: [No]

Justification: Section 3.3 presents Proposition 1 (non-absorption of delay parameters) with a proof sketch in Appendix A, grounded in linear delay-system theory. The argument establishes that τ_k is non-redundant and not absorbable into (κ_k, λ_k) , but is limited to the linearized, noise-free setting. A full structural-identifiability proof for the nonlinear stochastic system is explicitly left for future work. The answer is [No] because the proof is a sketch, not a complete formal proof.

Guidelines:

- The answer [N/A] means that the paper does not include theoretical results.
- All the theorems, formulas, and proofs in the paper should be numbered and cross-referenced.
- All assumptions should be clearly stated or referenced in the statement of any theorems.
- The proofs can either appear in the main paper or the supplemental material, but if they appear in the supplemental material, the authors are encouraged to provide a short proof sketch to provide intuition.
- Inversely, any informal proof provided in the core of the paper should be complemented by formal proofs provided in appendix or supplemental material.
- Theorems and Lemmas that the proof relies upon should be properly referenced.

4. Experimental result reproducibility

Question: Does the paper fully disclose all the information needed to reproduce the main experimental results of the paper to the extent that it affects the main claims and/or conclusions of the paper (regardless of whether the code and data are provided or not)?

Answer: [Yes]

Justification: The paper reports a completed 920-cell LLM-agent benchmark. The experimental setup is fully disclosed: IPD with 10% noise, 50-round cells, forced-defection perturbation at round 10, ablation conditions, opponent policies (TFT, GTFT, Grim, Random), provider/model assignment (OpenAI gpt-5-nano, Anthropic claude-haiku-4-5,

DeepSeek deepseek-v4-flash), 10 evaluation seeds per (condition, provider, opponent) cell, and frozen prompt templates. All 920 cells completed 50 rounds with balanced coverage; the bootstrap CIs in Table 3 use only complete cells.

Guidelines:

- The answer [N/A] means that the paper does not include experiments.
- If the paper includes experiments, a [No] answer to this question will not be perceived well by the reviewers: Making the paper reproducible is important, regardless of whether the code and data are provided or not.
- If the contribution is a dataset and/or model, the authors should describe the steps taken to make their results reproducible or verifiable.
- Depending on the contribution, reproducibility can be accomplished in various ways. For example, if the contribution is a novel architecture, describing the architecture fully might suffice, or if the contribution is a specific model and empirical evaluation, it may be necessary to either make it possible for others to replicate the model with the same dataset, or provide access to the model. In general, releasing code and data is often one good way to accomplish this, but reproducibility can also be provided via detailed instructions for how to replicate the results, access to a hosted model (e.g., in the case of a large language model), releasing of a model checkpoint, or other means that are appropriate to the research performed.
- While NeurIPS does not require releasing code, the conference does require all submissions to provide some reasonable avenue for reproducibility, which may depend on the nature of the contribution. For example
 - (a) If the contribution is primarily a new algorithm, the paper should make it clear how to reproduce that algorithm.
 - (b) If the contribution is primarily a new model architecture, the paper should describe the architecture clearly and fully.
 - (c) If the contribution is a new model (e.g., a large language model), then there should either be a way to access this model for reproducing the results or a way to reproduce the model (e.g., with an open-source dataset or instructions for how to construct the dataset).
 - (d) We recognize that reproducibility may be tricky in some cases, in which case authors are welcome to describe the particular way they provide for reproducibility. In the case of closed-source models, it may be that access to the model is limited in some way (e.g., to registered users), but it should be possible for other researchers to have some path to reproducing or verifying the results.

5. Open access to data and code

Question: Does the paper provide open access to the data and code, with sufficient instructions to faithfully reproduce the main experimental results, as described in supplemental material?

Answer: [No]

Justification: Code, data, and prompts are released through an anonymized public repository (https://anonymous.4open.science/r/proaction_operator-85D2/) preserving double-blind review. The reference simulator and verification suite (SymPy, Z3, mpmath checks) are self-contained and require no API keys. The LLM benchmark requires provider API access; the supplementary documents commands, dependencies, frozen prompts, checkpointing, seeds, and the filtering rule that final comparisons use only cells with 50 completed rounds.

Guidelines:

- The answer [N/A] means that paper does not include experiments requiring code.
- Please see the NeurIPS code and data submission guidelines (<https://neurips.cc/public/guides/CodeSubmissionPolicy>) for more details.
- While we encourage the release of code and data, we understand that this might not be possible, so [No] is an acceptable answer. Papers cannot be rejected simply for not including code, unless this is central to the contribution (e.g., for a new open-source benchmark).

- The instructions should contain the exact command and environment needed to run to reproduce the results. See the NeurIPS code and data submission guidelines (<https://neurips.cc/public/guides/CodeSubmissionPolicy>) for more details.
- The authors should provide instructions on data access and preparation, including how to access the raw data, preprocessed data, intermediate data, and generated data, etc.
- The authors should provide scripts to reproduce all experimental results for the new proposed method and baselines. If only a subset of experiments are reproducible, they should state which ones are omitted from the script and why.
- At submission time, to preserve anonymity, the authors should release anonymized versions (if applicable).
- Providing as much information as possible in supplemental material (appended to the paper) is recommended, but including URLs to data and code is permitted.

6. Experimental setting/details

Question: Does the paper specify all the training and test details (e.g., data splits, hyperparameters, how they were chosen, type of optimizer) necessary to understand the results?

Answer: [Yes]

Justification: No training or optimization is performed. The paper reports a completed experimental benchmark. The setting is fully specified: IPD environment, 50 rounds, 10% noise, forced-defection perturbation at round 10, four opponents, seven LLM ablations plus two numerical reductions, three providers, ten evaluation seeds, checkpointing, and exclusion of partial cells from comparable analyses. The main text includes a frozen-parameter provenance table; full details are in the code and supplementary material.

Guidelines:

- The answer [N/A] means that the paper does not include experiments.
- The experimental setting should be presented in the core of the paper to a level of detail that is necessary to appreciate the results and make sense of them.
- The full details can be provided either with the code, in appendix, or as supplemental material.

7. Experiment statistical significance

Question: Does the paper report error bars suitably and correctly defined or other appropriate information about the statistical significance of the experiments?

Answer: [No]

Justification: The paper reports 95% bootstrap confidence intervals (10k resamples) for cooperation rate, action volatility, and recovery delay across all nine conditions in Table 3. Pre-registered Holm–Bonferroni-corrected hypothesis tests for H1–H3 (Full- Γ vs. ReAct, Random- Γ , Collapse-NC) are reported in §5.1. Mann–Whitney U tests are reported for the No- $\mathcal{H}$ and No- $\mathcal{N}$ recovery-delay comparisons, with Cliff’s δ as effect size.

Guidelines:

- The answer [N/A] means that the paper does not include experiments.
- The authors should answer [Yes] if the results are accompanied by error bars, confidence intervals, or statistical significance tests, at least for the experiments that support the main claims of the paper.
- The factors of variability that the error bars are capturing should be clearly stated (for example, train/test split, initialization, random drawing of some parameter, or overall run with given experimental conditions).
- The method for calculating the error bars should be explained (closed form formula, call to a library function, bootstrap, etc.)
- The assumptions made should be given (e.g., Normally distributed errors).
- It should be clear whether the error bar is the standard deviation or the standard error of the mean.
- It is OK to report 1-sigma error bars, but one should state it. The authors should preferably report a 2-sigma error bar than state that they have a 96% CI, if the hypothesis of Normality of errors is not verified.

- For asymmetric distributions, the authors should be careful not to show in tables or figures symmetric error bars that would yield results that are out of range (e.g., negative error rates).
- If error bars are reported in tables or plots, the authors should explain in the text how they were calculated and reference the corresponding figures or tables in the text.

8. Experiments compute resources

Question: For each experiment, does the paper provide sufficient information on the computer resources (type of compute workers, memory, time of execution) needed to reproduce the experiments?

Answer: [Yes]

Justification: The reported LLM benchmark uses API-based inference through three providers and requires no specialized local hardware beyond a standard CPU workstation for orchestration and checkpointing. The paper/supplement reports provider usage, checkpointing, and approximate API-cost/latency diagnostics for the completed benchmark (per-cell amortized cost $\approx$ \$0.04–\$0.06 at the providers used; total $\sim$ \$45 across 920 cells).

Guidelines:

- The answer [N/A] means that the paper does not include experiments.
- The paper should indicate the type of compute workers CPU or GPU, internal cluster, or cloud provider, including relevant memory and storage.
- The paper should provide the amount of compute required for each of the individual experimental runs as well as estimate the total compute.
- The paper should disclose whether the full research project required more compute than the experiments reported in the paper (e.g., preliminary or failed experiments that didn't make it into the paper).

9. Code of ethics

Question: Does the research conducted in the paper conform, in every respect, with the NeurIPS Code of Ethics <https://neurips.cc/public/EthicsGuidelines>?

Answer: [Yes]

Justification: The paper does not use human subjects, scraped personal data, or deceptive practices. LLMs are used as policy-execution layers in a controlled IPD benchmark, and all outputs are experimental logs rather than personal data. The authors have reviewed the NeurIPS Code of Ethics and the paper includes a broader impacts discussion.

Guidelines:

- The answer [N/A] means that the authors have not reviewed the NeurIPS Code of Ethics.
- If the authors answer [No], they should explain the special circumstances that require a deviation from the Code of Ethics.
- The authors should make sure to preserve anonymity (e.g., if there is a special consideration due to laws or regulations in their jurisdiction).

10. Broader impacts

Question: Does the paper discuss both potential positive societal impacts and negative societal impacts of the work performed?

Answer: [Yes]

Justification: Section 6 (“Discussion and limitations”) discusses broader impacts: positive (interpretable subsystem-level regulation, falsifiable design enabling accountability) and negative (potential misuse for more persuasive deceptive agents). Mitigation strategies are discussed: interpretability at subsystem level and falsifiability by design.

Guidelines:

- The answer [N/A] means that there is no societal impact of the work performed.
- If the authors answer [N/A] or [No], they should explain why their work has no societal impact or why the paper does not address societal impact.

- Examples of negative societal impacts include potential malicious or unintended uses (e.g., disinformation, generating fake profiles, surveillance), fairness considerations (e.g., deployment of technologies that could make decisions that unfairly impact specific groups), privacy considerations, and security considerations.
- The conference expects that many papers will be foundational research and not tied to particular applications, let alone deployments. However, if there is a direct path to any negative applications, the authors should point it out. For example, it is legitimate to point out that an improvement in the quality of generative models could be used to generate Deepfakes for disinformation. On the other hand, it is not needed to point out that a generic algorithm for optimizing neural networks could enable people to train models that generate Deepfakes faster.
- The authors should consider possible harms that could arise when the technology is being used as intended and functioning correctly, harms that could arise when the technology is being used as intended but gives incorrect results, and harms following from (intentional or unintentional) misuse of the technology.
- If there are negative societal impacts, the authors could also discuss possible mitigation strategies (e.g., gated release of models, providing defenses in addition to attacks, mechanisms for monitoring misuse, mechanisms to monitor how a system learns from feedback over time, improving the efficiency and accessibility of ML).

11. Safeguards

Question: Does the paper describe safeguards that have been put in place for responsible release of data or models that have a high risk for misuse (e.g., pre-trained language models, image generators, or scraped datasets)?

Answer: [N/A]

Justification: The paper does not release a pre-trained model, dataset, or generative system. The contribution is a formal model (Γ) and reference simulator. No high-risk assets are released.

Guidelines:

- The answer [N/A] means that the paper poses no such risks.
- Released models that have a high risk for misuse or dual-use should be released with necessary safeguards to allow for controlled use of the model, for example by requiring that users adhere to usage guidelines or restrictions to access the model or implementing safety filters.
- Datasets that have been scraped from the Internet could pose safety risks. The authors should describe how they avoided releasing unsafe images.
- We recognize that providing effective safeguards is challenging, and many papers do not require this, but we encourage authors to take this into account and make a best faith effort.

12. Licenses for existing assets

Question: Are the creators or original owners of assets (e.g., code, data, models), used in the paper, properly credited and are the license and terms of use explicitly mentioned and properly respected?

Answer: [Yes]

Justification: All cited works (Friston 2017, Keramati–Gutkin 2014, Porcelli–Delgado 2017, etc.) are properly referenced in the bibliography. The NeurIPS 2026 LaTeX style file is used under its intended license. No proprietary datasets or models are used in this submission.

Guidelines:

- The answer [N/A] means that the paper does not use existing assets.
- The authors should cite the original paper that produced the code package or dataset.
- The authors should state which version of the asset is used and, if possible, include a URL.
- The name of the license (e.g., CC-BY 4.0) should be included for each asset.
- For scraped data from a particular source (e.g., website), the copyright and terms of service of that source should be provided.

- If assets are released, the license, copyright information, and terms of use in the package should be provided. For popular datasets, `paperswithcode.com/datasets` has curated licenses for some datasets. Their licensing guide can help determine the license of a dataset.
- For existing datasets that are re-packaged, both the original license and the license of the derived asset (if it has changed) should be provided.
- If this information is not available online, the authors are encouraged to reach out to the asset's creators.

13. New assets

Question: Are new assets introduced in the paper well documented and is the documentation provided alongside the assets?

Answer: [N/A]

Justification: No new dataset, model checkpoint, or benchmark is released in this submission. The reference implementation (`simulator.py`, `smoke_test.py`) is documented in the paper (Appendix A) and will be released with documentation upon acceptance.

Guidelines:

- The answer [N/A] means that the paper does not release new assets.
- Researchers should communicate the details of the dataset/code/model as part of their submissions via structured templates. This includes details about training, license, limitations, etc.
- The paper should discuss whether and how consent was obtained from people whose asset is used.
- At submission time, remember to anonymize your assets (if applicable). You can either create an anonymized URL or include an anonymized zip file.

14. Crowdsourcing and research with human subjects

Question: For crowdsourcing experiments and research with human subjects, does the paper include the full text of instructions given to participants and screenshots, if applicable, as well as details about compensation (if any)?

Answer: [N/A]

Justification: The paper does not involve crowdsourcing or research with human subjects. The Behavioral Fidelity Index compares agent behavior against published human distributions (Porcelli–Delgado 2017), not new human data collection.

Guidelines:

- The answer [N/A] means that the paper does not involve crowdsourcing nor research with human subjects.
- Including this information in the supplemental material is fine, but if the main contribution of the paper involves human subjects, then as much detail as possible should be included in the main paper.
- According to the NeurIPS Code of Ethics, workers involved in data collection, curation, or other labor should be paid at least the minimum wage in the country of the data collector.

15. Institutional review board (IRB) approvals or equivalent for research with human subjects

Question: Does the paper describe potential risks incurred by study participants, whether such risks were disclosed to the subjects, and whether Institutional Review Board (IRB) approvals (or an equivalent approval/review based on the requirements of your country or institution) were obtained?

Answer: [N/A]

Justification: No human subjects research is conducted in this paper. The BFI metric uses published aggregate distributions from prior studies, not individual-level data.

Guidelines:

- The answer [N/A] means that the paper does not involve crowdsourcing nor research with human subjects.
- Depending on the country in which research is conducted, IRB approval (or equivalent) may be required for any human subjects research. If you obtained IRB approval, you should clearly state this in the paper.
- We recognize that the procedures for this may vary significantly between institutions and locations, and we expect authors to adhere to the NeurIPS Code of Ethics and the guidelines for their institution.
- For initial submissions, do not include any information that would break anonymity (if applicable), such as the institution conducting the review.

16. Declaration of LLM usage

Question: Does the paper describe the usage of LLMs if it is an important, original, or non-standard component of the core methods in this research? Note that if the LLM is used only for writing, editing, or formatting purposes and does *not* impact the core methodology, scientific rigor, or originality of the research, declaration is not required.

Answer: [Yes]

Justification: LLMs are an important component of the core methodology: OpenAI, Anthropic, and DeepSeek models are used as the policy-execution layer of the agents, with each provider/model held fixed across conditions to isolate the Γ controller. AI-assisted tools, including an agentic IDE for code development, experiment orchestration, monitoring, and analysis support, were used under human review. Human authors reviewed all outputs and assume full responsibility for the paper, code, claims, and citations.

Guidelines:

- The answer [N/A] means that the core method development in this research does not involve LLMs as any important, original, or non-standard components.
- Please refer to our LLM policy in the NeurIPS handbook for what should or should not be described.